

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T05 · Fractions

With Answers

Questions with full worked solutions

■ 24 pages

FAMILY MAP

Fractions

F01

Add or subtract fractions and mixed numbers

4 Q | 12 marks

F02

Multiply or divide fractions and mixed numbers

6 Q | 18 marks

F03

Represent, simplify and compare fractions

1 Q | 4 marks

Working Unit 16

4MA1-2H Q3

ORIGINAL QUESTION

3 marks

3 Show that $7\frac{1}{3} - 3\frac{4}{7} = 3\frac{16}{21}$

(Total for Question 3 is 3 marks)

Worked Solution 16

Subtract mixed fractions

UNIT 1
3 marks**Convert**

$$7\frac{1}{3} - 3\frac{4}{7} = \frac{22}{3} - \frac{25}{7}.$$

Common denominator

$$\frac{154 - 75}{21} = \frac{79}{21} = 3\frac{16}{21}.$$

Final answer

$$3\frac{16}{21}$$

Question 18

4MA1-2H Q4

ORIGINAL QUESTION

3 marks

4 Show that $2\frac{4}{7} \times 3\frac{1}{9} = 8$

Worked Solution 18

Mixed-number multiplication

MODULAR UNIT 1

3 marks

Improper fractions

$$2\frac{4}{7} = 18/7, \quad 3\frac{1}{9} = 28/9.$$

Multiply

$$(18/7)(28/9) = 8.$$

Final answer

8

Question 20

4MA1-2H Q6 whole

ORIGINAL QUESTION UNIT 1

3 marks

6 Show that $3\frac{3}{7} \div 2\frac{2}{3} = 1\frac{2}{7}$

Worked solution 20

Show a mixed-number division identity

MODULAR UNIT 1

3 marks

whole – Show a mixed-number division identity

Convert to improper fractions

$$3\frac{3}{7} = \frac{24}{7}, \quad 2\frac{2}{3} = \frac{8}{3}.$$

Invert and multiply

$$\frac{24}{7} \div \frac{8}{3} = \frac{24}{7} \times \frac{3}{8} = \frac{72}{56} = \frac{9}{7}.$$

Convert back to a mixed number

$$\frac{9}{7} = 1\frac{2}{7}.$$

Final answer

1 2/7

Question 12

4MA1-2H Q1

ORIGINAL QUESTION

3 marks

1 Show that $4\frac{2}{3} \div 1\frac{1}{5} = 3\frac{8}{9}$

Worked Solution 12

Divide mixed numbers

MODULAR UNIT 1

3 marks

Convert

$$4\frac{2}{3} = \frac{14}{3}, \quad 1\frac{1}{5} = \frac{6}{5}.$$

Invert and multiply

$$\frac{14}{3} \div \frac{6}{5} = \frac{14}{3} \times \frac{5}{6} = \frac{70}{18} = \frac{35}{9}.$$

Final answer

$$\frac{35}{9} = 3\frac{8}{9}$$

Original question 17

4MA1-2H Q1 M01

MODULAR UNIT 1

3 marks

1 Show that $1\frac{5}{7} \times 2\frac{3}{16} = 3\frac{3}{4}$

Worked solution 17

Show the mixed-number product

MODULAR UNIT 1

3 marks

M01 – Show the mixed-number product**Convert to improper fractions**

$$1\frac{5}{7} = 12/7, \quad 2\frac{3}{16} = 35/16.$$

Multiply and cancel

$$\frac{12}{7} \times \frac{35}{16} = \frac{3 \times 5}{4} = \frac{15}{4}.$$

Convert to a mixed number

$$\frac{15}{4} = 3\frac{3}{4}.$$

Final answer

$$3\frac{3}{4}$$

Original question 17

4MA1-2H Q4 Q4

ORIGINAL QUESTION**3 marks**

4 Show that $5\frac{1}{3} - 2\frac{6}{7} = 2\frac{10}{21}$

Worked solution 17

Chapter: Fractions

MODULAR UNIT 1

3 marks

Q4 – Chapter: Fractions

Family: Add or subtract fractions and mixed numbers • Variant: Subtract fractions or mixed numbers

Method

$$5 \frac{1}{3} - 2 \frac{6}{7} = \frac{16}{3} - \frac{20}{7}.$$

*Why: Convert both mixed numbers to improper fractions.***Method**

$$\frac{16}{3} - \frac{20}{7} = \frac{112}{21} - \frac{60}{21} = \frac{52}{21}.$$

*Why: Use a common denominator of 21.***Method**

$$\frac{52}{21} = 2 \frac{10}{21}, \text{ which is the required result.}$$

*Why: Convert the improper fraction back to a mixed number.***Final answer****shown value:** $2 \frac{10}{21}$

Original question 20

4MA1-2H Q2

ORIGINAL QUESTION**3 marks**

2 Show that $6\frac{3}{4} \div 2\frac{4}{7} = 2\frac{5}{8}$

Worked solution 20

Chapter: Fractions

MODULAR UNIT 1

3 marks

Q2 – Chapter: Fractions

Family: Multiply or divide fractions and mixed numbers • Variant: Divide fractions or mixed numbers

Method $6\frac{3}{4} = \frac{27}{4}$ and $2\frac{4}{7} = \frac{18}{7}$.*Why: Convert both mixed numbers to improper fractions.***Method** $(\frac{27}{4}) \div (\frac{18}{7}) = (\frac{27}{4}) (\frac{7}{18}) = \frac{189}{72}$.*Why: Invert and multiply by the second fraction.***Method** $\frac{189}{72} = \frac{21}{8} = 2\frac{5}{8}$.*Why: Simplify and convert back to a mixed number.***Final answer**value: $2\frac{5}{8}$

3 Show that $4\frac{2}{3} + 3\frac{4}{5} = 8\frac{7}{15}$

Worked solution

January 2020 | Question 3 | 3 marks

Family: Add or subtract fractions and mixed numbers • Variant: Add fractions or mixed numbers

Method / working

1. $4\frac{2}{3} = \frac{14}{3}$ and $3\frac{4}{5} = \frac{19}{5}$.

2. $\frac{14}{3} + \frac{19}{5} = \frac{70}{15} + \frac{57}{15} = \frac{127}{15}$.

3. $\frac{127}{15} = 8\frac{7}{15}$.

Final answer: $8\frac{7}{15}$

2 Show that $2\frac{4}{7} \div 1\frac{1}{8} = 2\frac{2}{7}$

Worked solution

June 2021 | Question 2 | 3 marks

Family: Multiply or divide fractions and mixed numbers • Variant: Divide fractions or mixed numbers

Method / working

1. Convert mixed numbers: $2\frac{4}{7} = \frac{18}{7}$ and $1\frac{1}{8} = \frac{9}{8}$.
2. Divide by multiplying by the reciprocal: $(\frac{18}{7}) \times (\frac{8}{9}) = \frac{16}{7}$.
3. Convert $\frac{16}{7}$ back to the mixed number $2\frac{2}{7}$.

Final answer: $2\frac{2}{7}$

1 Show that $3\frac{5}{7} \div 1\frac{5}{8} = 2\frac{2}{7}$

(Total for Question 1 is 3 marks)

Answer reference | January 2023 | Question 1

Family: Add or subtract fractions and mixed numbers • Variant: Add fractions or mixed numbers

Question	Working	Answer	Mark	Notes
1	$\frac{26}{7}, \frac{13}{8}$ oe		3	M1 both fractions expressed as improper fractions, no need for $\div$ or $\times$ may be equivalent to those given eg $\frac{52}{14}, \frac{26}{16}$ etc. A student could invert $\frac{13}{8}$ and show multiplication - as shown in the 2nd M1, this mark is then implied.
	$\frac{26}{7} \times \frac{8}{13}$ oe or eg $\frac{208}{56} \div \frac{91}{56}$			M1 or for both fractions expressed as equivalent fractions with denominators that are a common multiple of 7 and 8 eg $\frac{208}{56} \div \frac{91}{56}$
	eg $\frac{26}{7} \times \frac{8}{13} = \frac{208}{91} = \frac{16}{7} = 2\frac{2}{7}$ or $\frac{26}{7} \times \frac{8}{13} = \frac{208}{91} = 2\frac{26}{91} = 2\frac{2}{7}$ or $\frac{26^2}{7} \times \frac{8}{13^1} = \frac{16}{7} = 2\frac{2}{7}$ or $\frac{208}{56} \div \frac{91}{56} = \frac{208}{91} = \frac{16}{7} = 2\frac{2}{7}$ or correct working to $\frac{16}{7}$ and writing $2\frac{2}{7} = \frac{16}{7}$ (usually on the first line of working) <i>working required</i>	shown		A1 dep on M2 NB: use of decimals scores no marks (unless used as a check)
Total 3 marks				

Worked solution

January 2023 | Question 1 | 3 marks

Family: Add or subtract fractions and mixed numbers • Variant: Add fractions or mixed numbers

Official final mark-scheme pages are embedded immediately after this question.

3 A box contains only

9 red bricks
43 blue bricks
and some yellow bricks

$\frac{7}{20}$ of the bricks are yellow bricks.

Each brick weighs 35 grams.

Work out the total weight of the yellow bricks.

..... grams

(Total for Question 3 is 4 marks)

Worked solution

November 2025 | Question 3 | 4 marks

Family: Represent, simplify and compare fractions • Variant: Fraction of a quantity or one number as a fraction of another

Method

There are $9+43=52$ red and blue bricks.

Let the total number of bricks be T .

Since $\frac{7}{20}$ of the bricks are yellow, the remaining fraction is

$$1 - \frac{7}{20} = \frac{13}{20}$$

So

$$\frac{13}{20}T = 52$$

$$T = 52 \times \frac{20}{13} = 80$$

Yellow bricks:

$$\frac{7}{20} \times 80 = 28$$

Each brick weighs 35 grams, so total yellow-brick weight is

$$28 \times 35 = 980$$

Answer: 980 grams